Product Name: GLAMIN inf.		Territory: EXP	Colour: ● BLACK ● DIE CUT	1. Draft 05.09. 2016 07.59 Uhr	Variable Data:
Type of Packaging: PIL	Dosage: 500ml				
Material number: M096575/02 EXP	2-D-Matrix Code M096575/02 EXP				
Pharma-Code (Laetus)	EAN Code:				
Dimension: 180 x 294 mm 1 PAGE	Font: Antique Olive Smallest Size: 8 Pt.			Operator: Christian Nagy +43(0) 34 52 72266-22	

Glamin

FRESENIUS KABI

Description
Solution for infusion.
Electrolyte-free, clear, colourless to slightly yellow solution of free amino acids and dipeptides for intravenous nutrition.

Composition
Content (per 1000 ml):

Alanine	16.00 g
Arginine	11.30 g
Aspartic acid	3.40 g
Glutamic acid	5.60 g
Glycyl-Glutamine H_2O	30.27 g
(corresp. to Glycine	10.27 g
corresp. to Glutamine 20.0 g)	
Glycyl-Tyrosine $2H_2O$	3.45 g
(corresp. to Glycine	0,94 g
corresp. to Tyrosine 2.28 g)	
Histidine	6.80 g
Isoleucine	5.60 g
Leucine	7.90 g
Lysine-Acetate	12.70 g
(corresp. to Lysine 9.0 g)	
Methionine	5.60 g
Phenylalanine	5.85 g
Proline	6.80 g
Serine	4.50 g
Threonine	5.60 g
Tryptophan	1.90 g
Valine	7.30 g
Citric acid	to pH 5.8
Water for injections	to 1000 ml

Amino acids/dipeptides	134 g/l
Total nitrogen	22.4 g/l
Energy content	2300 kJ (540 kcal)/l
Theoretical osmolarity	1040 mosm/l
Theoretical osmolality	1140 mosm/kg
Titration acidity to pH 7.4	approx. 60 mmol/l
pH	approx. 5.8
Density	1.0414 g/cm³

Clinical pharmacology
Glamin is an infusion solution for parenteral nutrition containing 18 essential and non-essential amino acids, three of which are in the form of the dipeptides glycyl-glutamine and glycyl-tyrosine.
The solution is suitable to support protein synthesis and to improve nitrogen balance during intravenous nutrition. In order to ensure optimal utilization of the infused amino acids and dipeptides, the patient´s requirements of energy (carbohydrates, fat), electrolytes, trace elements and vitamins should be covered.

Indications
Glamin provides amino acids as part of parenteral nutrition therapy, when oral or enteral nutrition is impossible, insufficient or contraindicated, especially in patients with a moderate to severe catabolic status.
In parenteral nutrition regimens amino acid solutions should always be administered in combination with appropriate energy-supplying infusion solutions.

Contraindications
Patients with inborn errors of amino acid metabolism (e.g. phenylketonuria), severe liver failure and severe renal failure.
General contraindicatons of parenteral nutrition are: unstable life-threatening circulatory conditions (shock), metabolic acidosis, insufficient cellular oxygen supply, hyperhydration, hyponatremia, hypokalemia, hyperlactatemia, increased serum osmolarity, pulmonary oedema, decompensated cardiac insufficiency and known hypersensitivity to any of the ingredients.

Caution
Amino acid solutions should not be used as carrier solutions for drugs.

Glamin may only be mixed with other solutions where compatibility is documented. Serum electrolytes, serum osmolarity, water balance, acid-base status as well as liver function tests (alkaline phosphatase, GPT, GOT) should be monitored.

Use in pediatric patients
Glamin is not indicated for use in children below the age of 2 years since its composition is not adapted to the requirements of these patients. For older children experience is lacking, the use of Glamin can therefore not be recommended.

Pregnancy and lactation
No human data are available on the use of Glamin during pregnancy and lactation. The use of Glamin during pregnancy and lactation should be subject to a benefit-risk evaluation.
However, an evaluation of experimental animal studies (embryotoxicity study in rabbit) does not indicate direct or indirect harmful effect with respect to reproduction.

Adverse events
Not to be expected, if used as directed.

Drug interactions
Not investigated, however, no interactions are known to date.

Dosage
The dosage depends on the amino acid requirements.
Generally, 1-2 g amino acids/dipeptides (corresp. to 0.17-0.34 g N) per kg body weight per day are recommended. This corresponds to 7-14 ml Glamin/kg body weight/day or to 500-1000 ml Glamin/day for a patient weighing 70 kg.
Recommended infusion rate: 0.6-0.7 ml (corresp. to 0.08-0.09 g amino acids/dipeptides)/kg body weight/ hour. This corresponds to 500 ml in 10-12 hours or 1000 ml in 20-24 hours for a patient weighing 70 kg.
Dosage to be adjusted individually for patients with renal or liver diseases.

Administration
Intravenous infusion. Glamin should be administered by the central venous route due to its osmolarity above 800 mosm/l.
Infusion may be continued for as long as required by the patient´s clinical condition. No experience is available so far for administration over more than 2 weeks.
Use only clear solutions in intact containers.
To achieve a complete parenteral nutrition regimen, Glamin should be administered in combination with carbohydrates and/or fat as well as electrolytes, trace elements and vitamins.

Overdose
When infusion rates exceed the recommended maximum rate, signs of intolerance may occur: nausea, vomiting, flushing, sweating in combination with renal excretion of amino acids and dipeptides.
Therapy if symptoms of overdose occur: Reduce infusion rate or, if necessary, interrupt infusion.

Compatibility
For the following mixture with Glamin compatibility is documented:
1000 ml Glamin with 20% fat emulsion (up to 1000 ml Intralipid 20%*), up to 1000 ml glucose 40%, 80 mmol NaCl, 5 mmol $CaCl_2$, 60 mmol KCl, 3.5 mmol Mg-L-hydrogenglutamate, phosphate supplement (15 ml Addiphos*), trace elements (10 ml Addamel N*/Addel N*), fat soluble vitamins (10 ml Vitalipid N Adult*) and water soluble vitamins (1 vial Soluvit N*).
*) used for compatibility testing

Additions should be performed aseptically immediately before the start of the infusion. Discard any residual contents.

Storage
Do not store above +25°C.

Pack sizes
Bottles of 250 ml, 500 ml and 1000 ml.

Manufactured by: **Fresenius Kabi** Austria GmbH, Graz, Austria for **Fresenius Kabi AB**, Uppsala, Sweden

M096575/02 EXP

APPROVED
BPM/PR-Planung, Graz
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Datum: ____________ Unterschrift: ______________